DARIO TRINCHERO

PhD Candidate in Mathematics

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PROFILE

I am an early-career mathematical physicist working in low-dimensional topology and Chern–Simons theory. I aspire toward a position that combines research with lecturing and supervision.

I am passionate about teaching: I have tutored undergraduate mathematics since 2018 and lectured a large first-year class in 2024. I see teaching as a means to give back the guidance and opportunities I was privileged to receive in my own education.

EDUCATION

Doctorate in Mathematics

Jan 2023 – Nov 2026 | Stellenbosch University, ZA

Thesis: *Coherent path states and coloured Jones asymptotics in Chern–Simons theory*

Master's in Theoretical Physics

Feb 2021 – Dec 2022 | Stellenbosch University, ZA

Thesis: *Pinhole interference in 3D fuzzy space*

Courses: Quantum information, relativity, solid-state physics

Honours in Mathematics

Feb 2020 – Dec 2020 | Stellenbosch University, ZA

Mini Thesis: *Generalised Feynman formula for Ising model*

Courses: Quantum field theory, (functional) analysis

Bachelor's in Theoretical Physics

Feb 2017 – Dec 2019 | Stellenbosch University, ZA

Majors: Mathematics & physics (+8 extra modules)

Cambridge International AS- & A-Levels

Jan 2015 – Nov 2016 | Somerset College, ZA

A-Levels: Mathematics, physics, computer science, chemistry

RESEARCH

Journal Articles

- **D. Trinchero** and F. G. Scholtz, "Pinhole interference in three-dimensional fuzzy space," *Annals of Physics*, vol. 450, p. 169 224, Mar. 2023. DOI: 10.1016/j.aop.2023.169224

Theses

- **D. Trinchero**, "Pinhole interference in three-dimensional fuzzy space," MSc thesis, Stellenbosch University, Mar. 2023. HDL: 10019.1/126944.

HIGHLIGHTS

- First-author paper in the *Annals of Physics* (2023)
- Presented research at Syracuse University and Notre Dame
- Teaching award: chosen by students as their most impactful lecturer (2025)
- Dean's Medal for top faculty average; all degrees *cum laude*

SKILLS & INTERESTS

Research interests

Chern–Simons theory | TQFT | Skein theory | Theta functions | Geometric quantisation | Quantum groups | Low-dimensional topology

Further interests

Quantum field theory | Lie theory | Category theory | Quantum computing

Programming (since 2011)

Python | C/C++ | Java | TypeScript | React | x86 Assembly | Ruby | Octave | MATLAB | Mathematica | Rust (learning) | TEX | Unix

REFEREES

Prof. Bruce Bartlett (Hons & PhD supervisor)
bbartlett@sun.ac.za
Mathematics department
Stellenbosch University, South Africa

Prof. Frederik Scholtz (MSc supervisor)
fgs@sun.ac.za
Physics department
Stellenbosch University, South Africa

Dr Johannes Kriel (MSc examiner)
hkriel@sun.ac.za
Physics department
Stellenbosch University, South Africa

Talks & Posters

- **D. Trinchero**, *Geometry of theta functions in TQFT*, Mathematics Department Seminar, Stellenbosch University, Jun. 2025.
- **D. Trinchero**, *A new isomorphism between the spaces of skeins & holomorphic sections in Chern–Simons theory*, Poster, CMND, Notre Dame, Jun. 2024.
- **D. Trinchero**, *Introduction to quantum groups*, Stellenbosch University, Oct. 2023. DOI: 10.5281/zenodo.10000346
- **D. Trinchero**, *Tour of knots & theta functions*, Stellenbosch University, Oct. 2023. DOI: 10.5281/zenodo.10047936
- **D. Trinchero**, *Pinhole interference in 3D fuzzy space*, Syracuse University, Sep. 2023. [Online]. Available: https://video.syr.edu/media/t/1_74dkljgj
- **D. Trinchero**, *Computing by collapsing*, Cosmic Conversations, Stellenbosch University, Apr. 2022. DOI: 10.5281/zenodo.8228648

Complete list: https://dariotrinchero.github.io/talks/

TEACHING

Junior Mathematics Lecturer

Feb 2024 – Jul 2024 — Stellenbosch University, ZA

I lectured calculus part-time to a class of around 220, and set tutorial & exam questions. My students chose me for a ROCTAS teaching award in 2025.

Mathematics Course Assistant

Feb 2018 – present — Stellenbosch University, ZA

I tutored these undergraduate classes, sometimes in charge of marking:

- *3rd year:* Algebra, real analysis, Fredholm theory
- *2nd year:* Linear algebra, advanced calculus, analysis
- *1st year:* Linear algebra, calculus

Private Tutoring

Aug 2017 – Mar 2018 — Independent, *ad hoc*

Grade 11–12 mathematics and physics, both national curriculum & advanced programme mathematics.

EMPLOYMENT

AI Safety Research Fellowship

Nov 2026 – Feb 2027 — PIBBSS, Cape Town, ZA

I was selected for the PIBBSS AI safety fellowship to research computational learning theory, mentored by Vanessa Kosoy.

Software Development Engineer Intern

Nov 2019 – Feb 2020
Dec 2020 – Feb 2021 — Amazon Web Services, ZA

I interned twice for *Amazon Web Services* (AWS) where I worked on *Elastic Compute Cloud* (EC2). I built internal tools for my team using TypeScript & Java on AWS (e.g. Lambda, DynamoDB, S3).

ACHIEVEMENTS

2023–2026: PhD

- ROCTAS Award: students choose their most impactful lecturer (2025)
- Bursary: National Research Foundation

2021–2022: MSc

- 100% for 1 of 3 modules; median mark of 97%; 89% for thesis
- Bursaries: Harry Crossley & Skye Foundations

2020: BSc Hons

- 100% for 1 of 11 modules; median mark of 94%; 95% for mini thesis
- Dean's Medal: top faculty average over 4 years
- Perimeter Institute: attended summer program
- Bursary: National Research Foundation

2017–2019: BSc

- 100% for 16 of 37 modules; median mark of 99%
- Rector's Award: among top students of faculty
- Rubbi Book Prize: top in Mathematics (2018, 2019)
- Top Computer Science & Applied Mathematics first year
- First Year Achievement Awards: among top first year students
- Team contests (2018): 1st in South African Mathematical Modelling Contest; 2nd in *SANReN* Cyber Security Challenge
- Bursary: Stellenbosch Merit Award (2018, 2019)

2015–2016: A-levels

- Dux scholar: Somerset College (Cambridge)
- Top marks for AS-levels: Mathematics (globally), Chemistry (South Africa)

PERSONAL PROJECTS

- **isiXhosa** (since 2025): self-studying this language, widely spoken in my home city of Cape Town.
- **Philosophy** (since 2021): working through papers & textbooks; I gave a talk on a famous (alleged) mathematical proof of God.
- **Game development** (since 2019): annual *Global Game Jam*, and several prototypes of my own, such as a web version of Hive.
- **Coding challenges** (since 2015): solving *Project Euler* problems.

Hobbies

I enjoy reading, cooking, trail-running, rock-climbing, and tabletop games, especially the abstract strategy game Hive.